

A low-fat vegan diet and a conventional diabetes diet in the treatment of type 2 diabetes: a randomized, controlled, 74-wk clinical trial

Background: Low-fat vegetarian and vegan diets are associated with weight loss, increased insulin sensitivity, and improved cardiovascular health. Objective: We compared the effects of a low-fat vegan diet and conventional diabetes diet recommendations on glycemia, weight, and plasma lipids. Design: Free-living individuals with type 2 diabetes were randomly assigned to a low-fat vegan diet (n = 49) or a diet following 2003 American Diabetes Association guidelines (conventional, n = 50) for 74 wk. Glycated hemoglobin (Hb A1c) and plasma lipids were assessed at weeks 0, 11, 22, 35, 48, 61, and 74. Weight was measured at weeks 0, 22, and 74. Results: Weight loss was significant within each diet group but not significantly different between groups (−4.4 kg in the vegan group and −3.0 kg in the conventional diet group, $P = 0.25$) and related significantly to Hb A1c changes ($r = 0.50$, $P = 0.001$). Hb A1c changes from baseline to 74 wk or last available values were −0.34 and −0.14 for vegan and conventional diets, respectively ($P = 0.43$). Hb A1c changes from baseline to last available value or last value before any medication adjustment were −0.40 and 0.01 for vegan and conventional diets, respectively ($P = 0.03$). In analyses before alterations in lipid-lowering medications, total cholesterol decreased by 20.4 and 6.8 mg/dL in the vegan and conventional diet groups, respectively ($P = 0.01$); LDL cholesterol decreased by 13.5 and 3.4 mg/dL in the vegan and conventional groups, respectively ($P = 0.03$). Conclusions: Both diets were associated with sustained reductions in weight and plasma lipid concentrations. In an analysis controlling for medication changes, a low-fat vegan diet appeared to improve glycemia and plasma lipids more than did conventional diabetes diet recommendations. Whether the observed differences provide clinical benefit for the macro- or microvascular complications of diabetes remains to be established. This trial was registered at clinicaltrials.gov as NCT00276939.